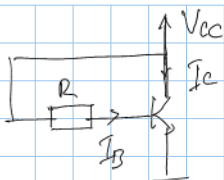
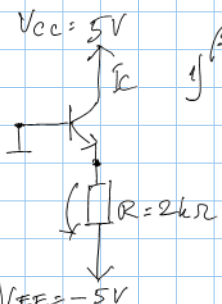


1)  $\beta = 300$; $I_S = 24 \text{ fA}$; $I_C = 10 \text{ mA}$
 $V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S}\right) = 25,8 \text{ mV} \cdot \ln\left(\frac{10 \text{ mA}}{24 \text{ fA}}\right) = 690,29 \text{ mV}$
 Maschengleichung: $V_{CC} = R \cdot I_B + V_{BE0}$
 $R = \frac{V_{CC} - V_{BE0}}{I_B} = \frac{V_{CC} - V_{BE0}}{I_C / \beta} = \frac{5 \text{ V} - 0,69029 \text{ V}}{10 \text{ mA} / 300} = 129,29 \text{ k}\Omega$

2)  $\beta = 300$; $I_S = 24 \text{ fA}$; $R = 5 \text{ k}\Omega$
 Konstant spannungsabfall-Modell $V_{BE0} = 0,7 \text{ V}$
 $V_{BE} = V_B - V_E \Rightarrow V_E = V_B - V_{BE} = -V_{BE} = -0,7 \text{ V}$
 $V_R = I_E \cdot R = V_E - V_{EE} = -V_{BE} - V_{EE}$
 $I_E = \frac{-V_{BE} - V_{EE}}{R} \Rightarrow I_C = \alpha \cdot I_E = \frac{\alpha \cdot (-V_{BE} - V_{EE})}{R}$
 $I_C = \frac{300}{1+300} \cdot \frac{(-0,7 - (-5 \text{ V}))}{2 \text{ k}\Omega} = 2,143 \text{ mA}$

2) Iterationsverfahren
 $I_C(n) = \frac{\alpha \cdot (-V_{BE}(n) - V_{EE})}{R} \quad \& \quad V_{BE}(n+1) = V_T \ln\left(\frac{I_C(n)}{I_S}\right)$

• Startwert $V_{BE}(0) = 0,7 \text{ V}$
 $I_C(0) = \frac{\alpha \cdot (-V_{BE}(0) - V_{EE})}{R} = \frac{300}{1+300} \cdot \frac{(-0,7 \text{ V} - (-5 \text{ V}))}{2 \text{ k}\Omega} = 2,14286 \text{ mA}$

• $V_{BE}(1) = V_T \cdot \ln\left(\frac{I_C(0)}{I_S}\right) = 25,8 \text{ mV} \cdot \ln\left(\frac{2,14286 \text{ mA}}{24 \text{ fA}}\right) = 650,5498 \text{ mV}$
 $I_C(1) = \frac{\alpha \cdot (-V_{BE}(1) - V_{EE})}{R} = \frac{300}{1+300} \cdot \frac{(-0,6505498 - (-5 \text{ V}))}{2 \text{ k}\Omega} = 2,1675 \text{ mA}$

Abbruchkriterium: $|V_{BE}(1) - V_{BE}(0)| = |650,5498 \text{ mV} - 700 \text{ mV}| = 49,45 \text{ mV} > \epsilon_V = 10 \mu\text{V}$
 $|I_C(1) - I_C(0)| = |2,1675 \text{ mA} - 2,14286 \text{ mA}| = 24,6 \mu\text{A} > \epsilon_I = 10 \text{ nA}$

• $V_{BE}(2) = 25,8 \text{ mV} \cdot \ln\left(\frac{2,1675 \text{ mA}}{24 \text{ fA}}\right) = 650,8448 \text{ mV}$
 $I_C(2) = \frac{300}{1+300} \cdot \frac{(-650,8448 \text{ mV} - (-5 \text{ V}))}{2 \text{ k}\Omega} = 2,167353 \text{ mA}$

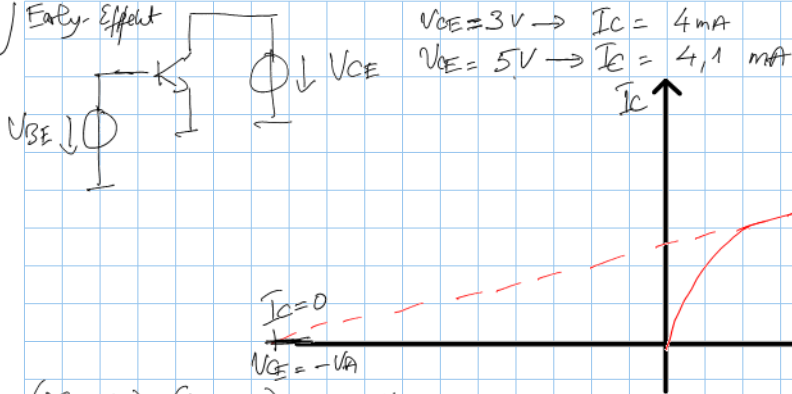
Abbruchkriterium: $|650,8448 \text{ mV} - 650,5498 \text{ mV}| = 295 \mu\text{V} > \epsilon_V = 10 \mu\text{V}$
 $|2,167353 \text{ mA} - 2,1675 \text{ mA}| = 147 \text{ nA} > \epsilon_I = 10 \text{ nA}$

• $V_{BE}(3) = 25,8 \text{ mV} \cdot \ln\left(\frac{2,167353 \text{ mA}}{24 \text{ fA}}\right) = 650,843 \text{ mV} \rightarrow \text{Konvergiert}$
 $I_C(3) = \frac{300}{1+300} \cdot \frac{(-650,843 \text{ mV} - (-5 \text{ V}))}{2 \text{ k}\Omega} = 2,167354 \text{ mA} \rightarrow \text{Konvergiert}$

Abbruchkriterium: $|650,843 \text{ mV} - 650,8448 \text{ mV}| = 1,75 \mu\text{V} < \epsilon_V = 10 \mu\text{V}$
 $|2,167353 \text{ mA} - 2,167354 \text{ mA}| = 0,872 \text{ nA} < \epsilon_I = 10 \text{ nA}$

Lösung: $I_C = 2,167 \text{ mA}$

3) Early-Effekt



$$(x_1, y_1) (x_2, y_2) \Rightarrow y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}; (x_1, y_1) \Rightarrow y_1 = \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1 + c \Rightarrow c = y_1 - \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1$$

$$\Leftrightarrow y = \frac{y_2 - y_1}{x_2 - x_1} \cdot x + y_1 - \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1$$

$$y = \frac{(y_2 - y_1)(x - x_1)}{x_2 - x_1} + y_1 \Leftrightarrow \frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Formel für lineare Gleichung mit zwei Punkten.

$$(V_{CE}, I_C): (3V, 4mA), (5V, 4.1mA)$$

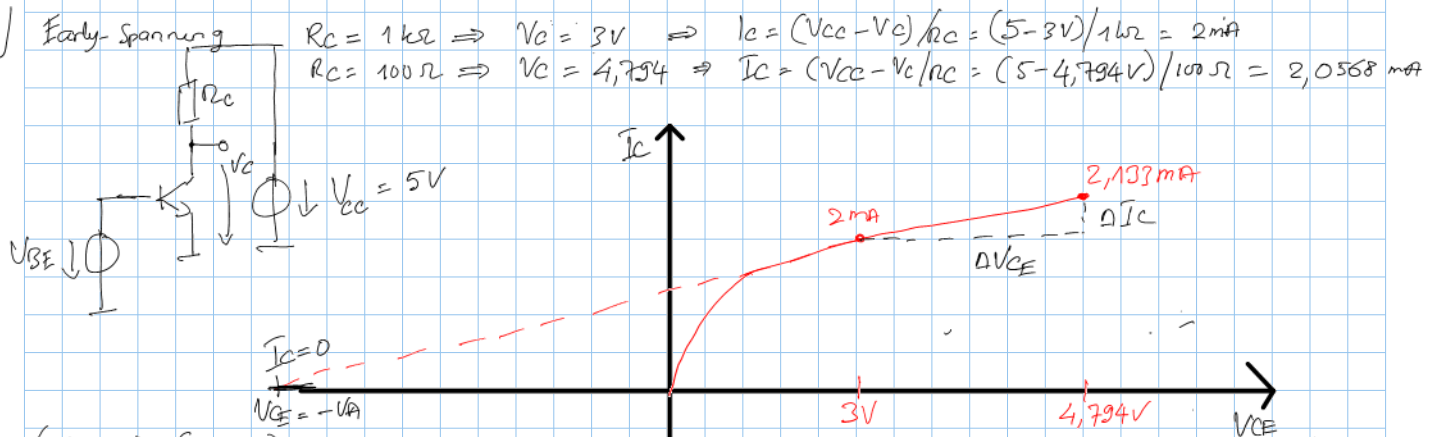
$$\Leftrightarrow \frac{I_C - 4mA}{4.1mA - 4mA} = \frac{V_{CE} - 3V}{5V - 3V}; \text{ für } I_C = 0 \Rightarrow \frac{-4mA}{0.1mA} \cdot 2 + 3 = V_{CE}$$

$$\Leftrightarrow V_{CE} = -77V \Rightarrow V_A = 77V$$

$$I_C = I_S \cdot \exp\left(\frac{V_{BE}}{V_T}\right) \cdot \left(1 + \frac{V_{CE}}{V_A}\right) \Rightarrow V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S} \cdot \frac{1}{1 + V_{CE}/V_A}\right)$$

$$V_{BE} = 25.8mV \cdot \ln\left(\frac{4mA}{24\mu A} \cdot \frac{1}{1 + 3V/77}\right) = 665.67mV$$

4) Early-Spannung



$$(x_1, y_1) (x_2, y_2) \Rightarrow y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}; (x_1, y_1) \Rightarrow y_1 = \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1 + c \Rightarrow c = y_1 - \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1$$

$$\Leftrightarrow y = \frac{y_2 - y_1}{x_2 - x_1} \cdot x + y_1 - \frac{y_2 - y_1}{x_2 - x_1} \cdot x_1$$

$$y = \frac{(y_2 - y_1)(x - x_1)}{x_2 - x_1} + y_1 \Leftrightarrow \frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

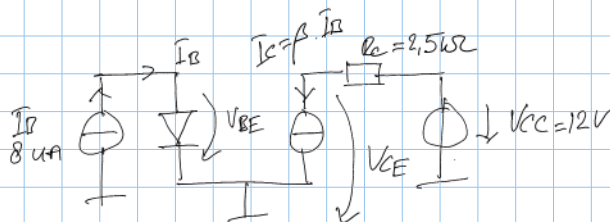
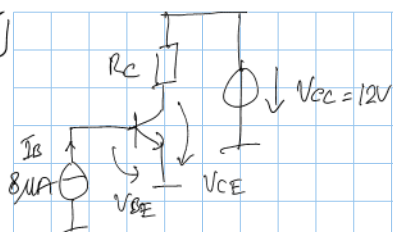
Formel für lineare Gleichung mit zwei Punkten.

$$(V_{CE}, I_C): (3V, 2mA), (4.794V, 2.0568mA)$$

$$\Leftrightarrow \frac{I_C - 2mA}{2.0568mA - 2mA} = \frac{V_{CE} - 3V}{4.794V - 3V}; \text{ für } I_C = 0 \Rightarrow \frac{-2mA}{0.0568mA} \cdot 1.794V + 3V = V_{CE}$$

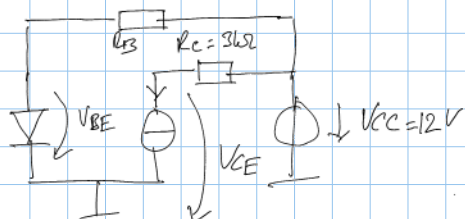
$$\Leftrightarrow V_{CE} = -60.169V \Rightarrow V_A = 60.169V$$

5)



$$I_C = \beta \cdot I_B = 300 \cdot 8 \mu A = 2.4 \text{ mA}; \quad I_C = I_S \cdot \exp\left(\frac{V_{BE}}{V_T}\right) \Rightarrow V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S}\right) = 25.8 \text{ mV} \cdot \ln\left(\frac{2.4 \text{ mA}}{24 \text{ fA}}\right) = 653.47 \text{ mV}$$

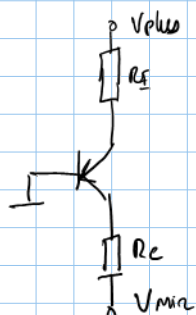
$$V_{CE} = V_{CC} - I_C \cdot R_C = 12 \text{ V} - 2.4 \text{ mA} \cdot 2.5 \text{ k}\Omega = 6 \text{ V} > V_{CEsat} \Rightarrow \text{Transistor ist im aktiven Bereich}$$



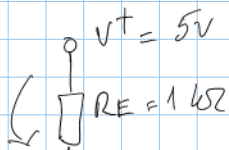
$$V_{CC} = I_B R_B + V_{BE}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{8 \mu A} = 1.4125 \text{ M}\Omega$$

6)

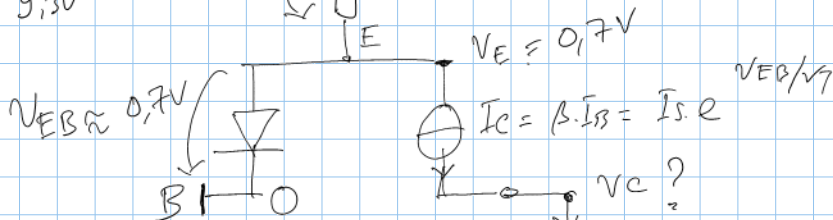


$$V_{CE} = 10 \text{ V} - 0.7 \text{ V} = 9.3 \text{ V}$$



$$I_C = ?$$

$$I_E = ?$$



$$I_E = \frac{V^+ - V_{BE}}{R} = \frac{5 - 0.7 \text{ V}}{1 \text{ k}\Omega} = 4.3 \text{ mA}$$

$$I_C = \frac{\beta}{\beta + 1} \cdot I_E = \frac{95}{95 + 1} \cdot 4.3 \text{ mA} = 4.2552 \text{ mA}$$

$$V_C = R_C \cdot I_C + V^- = 500 \Omega \cdot 4.2552 \text{ mA} + (-5 \text{ V}) = -2.8724 \text{ V}$$

$$V_{EC} = V_E - V_C = 0.7 \text{ V} - (-2.8724 \text{ V}) = 3.5724 \text{ V} > V_{CEsat}$$

⇒ Transistor wird im aktiven Bereich betrieben

Aktiver Bereich

$$V_{EC} \geq V_{CEsat} = 0.2 \text{ V}$$

$$V_E - V_C \geq 0.2 \text{ V}$$

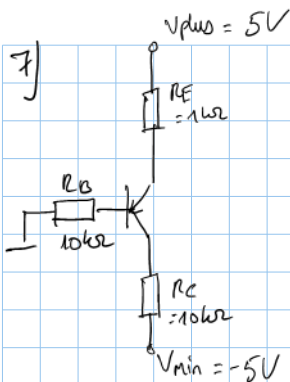
$$V_E - 0.2 \text{ V} \geq V_C$$

$$0.7 - 0.2 \text{ V} = 0.5 \text{ V} \geq V_C = I_C \cdot R_C + V^-$$

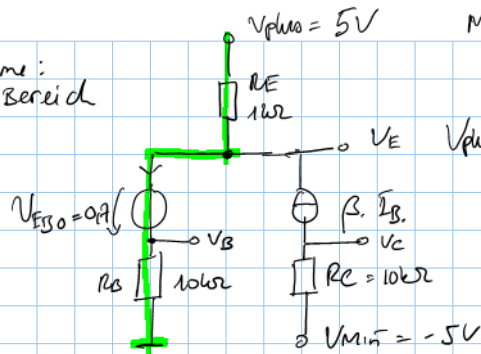
$$\frac{0.5 \text{ V} - V^-}{I_C} \geq R_C$$

$$\frac{0.5 \text{ V} - (-5 \text{ V})}{4.2552 \text{ mA}} \geq R_C$$

$$1.293 \text{ k}\Omega \geq R_C$$



Annahme:
Aktiver Bereich



Maschengleichung:

$$V_{plus} = R_E \cdot I_E + V_{BE0} + R_B \cdot I_B$$

$$V_{plus} - V_{BE0} = (R_E \cdot (1 + \beta) + R_B) I_B$$

$$I_B = \frac{V_{plus} - V_{BE0}}{R_E \cdot (1 + \beta) + R_B}$$

$$I_B = \frac{5V - 0.7V}{1k\Omega \cdot (1 + 30) + 10k\Omega}$$

$$I_B = 104.878 \mu A$$

$$\Rightarrow V_E = V_{plus} - R_E \cdot I_E = 5V - 1k\Omega \cdot (1 + 30) \cdot 104.878 \mu A = 1.749 V$$

$$I_C = \beta \cdot I_B = 30 \cdot 104.878 \mu A = 3.146 mA$$

$$\Rightarrow V_C = V_{min} + R_C \cdot I_C = -5V + 10k\Omega \cdot 3.146 mA \gg V_E$$

Widerspruch $\Rightarrow$ Transistor ist nicht im aktiven Bereich da V_{EC} nicht $> V_{ECsat}$

$\Rightarrow$ Annahme: Sättigungsbereich

$$V_E = V_B + V_{BE0} = V_B + 0.7V$$

$$V_C = V_E - V_{ECsat} = V_B + 0.7V - 0.2V = V_B + 0.5V$$

$$I_E = \frac{V_{plus} - V_E}{R_E} = \frac{5V - V_B - 0.5V}{1k\Omega} = 4.3 mA - \frac{V_B}{1k\Omega}$$

$$I_B = \frac{V_B}{10k\Omega}$$

$$I_C = \frac{V_C - V_{min}}{R_C} = \frac{V_B + 0.5V - (-5V)}{10k\Omega} = \frac{V_B}{10k\Omega} + 0.55 mA$$

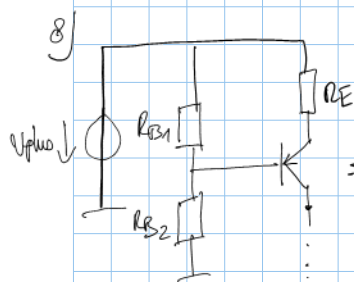
$$I_E = I_B + I_C \Rightarrow 4.3 mA - \frac{V_B}{1k\Omega} = \frac{V_B}{10k\Omega} + \frac{V_B}{10k\Omega} + 0.55 mA$$

$$3.75 mA = \frac{V_B}{5k\Omega} + \frac{5V_B}{5k\Omega} \Rightarrow 6 V_B = 3.75 mA \cdot 5k\Omega \Rightarrow V_B = 3.125 V$$

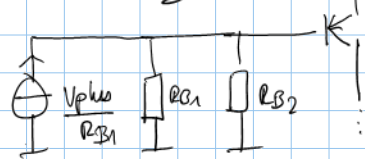
$$\Rightarrow V_E = 3.125V + 0.7V = 3.825V; V_C = 3.125V + 0.5V = 3.625V; I_B = \frac{3.125V}{10k\Omega} = 312.5 \mu A$$

$$I_C = \frac{3.125V}{10k\Omega} + 0.55 mA = 862.5 \mu A; I_E = I_B + I_C = 312.5 \mu A + 862.5 \mu A = 1.175 mA$$

$$\beta = \frac{I_C}{I_B} = \frac{862.5 \mu A}{312.5 \mu A} = 2.76 \ll \beta = 30 \Rightarrow \text{Transistor ist gesättigt.}$$

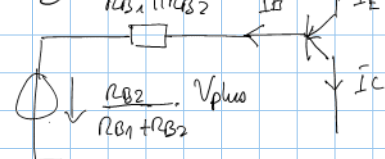


Quellenwandlung



$\leftarrow (1 + \beta) \cdot I_B$

Quellenwandlung



$$V_{plus} = R_E \cdot I_E + V_{BE0} + R_{B1} \parallel R_{B2} \cdot I_B + \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot V_{plus}$$

$$I_B = \frac{(V_{plus} - V_{BE0} - \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot V_{plus})}{R_E \cdot (1 + \beta) + R_{B1} \parallel R_{B2}}$$

$$I_B = \frac{(5V - 0.7V - \frac{300k}{300k + 300k} \cdot 5V)}{250 \cdot (95 + 1) + 300k \parallel 300k} = 10.845 \mu A$$

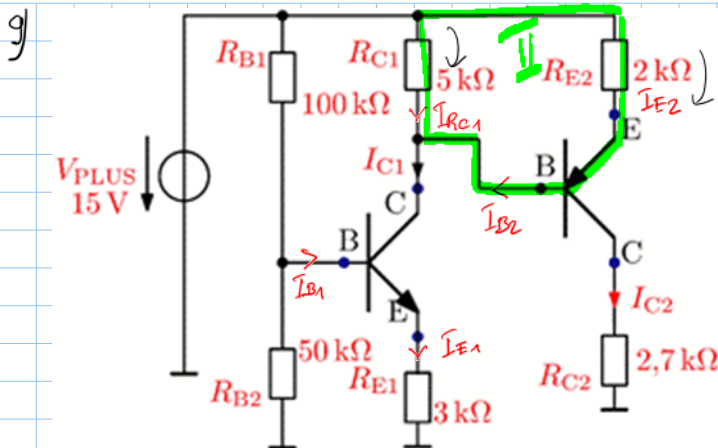
$$I_C = \beta \cdot I_B = 0.983 mA; I_E = I_B + I_C = 0.993 mA$$

$$V_E = V_{plus} - R_E \cdot I_E = 5V - 250\Omega \cdot 0.993 mA = 4.752V$$

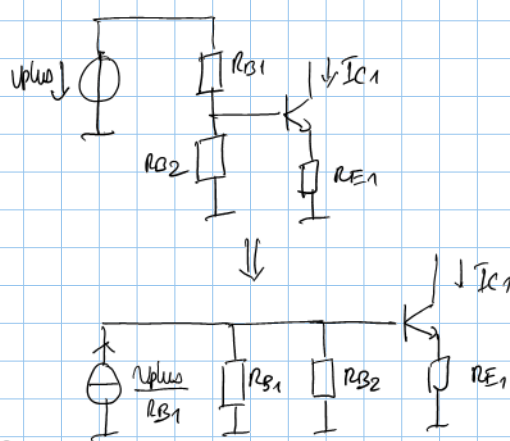
$$V_C = R_C \cdot I_C = 2k\Omega \cdot 0.983 mA = 1.966V$$

$$V_{EC} = V_E - V_C = 4.752V - 1.966V = 2.786V > V_{ECsat}$$

$\Rightarrow$ Transistor ist im aktiven Betriebsbereich



Methode der Quellenwandlung:



★ Maschengleichung I

$$- \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot V_{PLUS} + (R_{B1} \parallel R_{B2}) I_{B1} + V_{BE0} + R_{E1} \cdot I_{E1} = 0$$

$$\Rightarrow (R_{B1} \parallel R_{B2} + R_{E1}(1+\beta)) I_{B1} = \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot V_{PLUS} - V_{BE0}$$

$$I_{B1} = \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot \frac{V_{PLUS} - V_{BE0}}{\frac{R_{B1} \cdot R_{B2}}{R_{B1} + R_{B2}} + R_{E1}(1+\beta)}$$

$$I_{B1} = \frac{50k}{100k + 50k} \cdot \frac{15V - 0,7}{\frac{50k \cdot 100k}{100k + 50k} + 3k\Omega(1+100)} = 12,785 \mu A \Rightarrow I_{C1} = \beta \cdot I_{B1} = 100 \cdot 12,785 \mu A$$

$$I_{C1} = 1,2875 mA$$

$$I_{E1} = I_{B1} + I_{C1} = 1,3 mA$$

$$I_{C1} = I_{RC1} + I_{B2} \Rightarrow I_{RC1} = I_{C1} - I_{B2}$$

★ Maschengleichung II: $-V_{RC1} + V_{RE2} + V_{BE0} = -R_{C1} \cdot I_{RC1} + R_{E2} \cdot I_{E2} + V_{BE0} = 0$

$$\Rightarrow -R_{C1}(I_{C1} - I_{B2}) + R_{E2} \cdot (1+\beta) \cdot I_{B2} + V_{BE0} = 0$$

$$(R_{C1} + R_{E2}(1+\beta)) I_{B2} = R_{C1} \cdot I_{C1} - V_{BE0}$$

$$I_{B2} = \frac{R_{C1} \cdot I_{C1} - V_{BE0}}{R_{C1} + R_{E2}(1+\beta)} = \frac{5k\Omega \cdot 1,2875 mA - 0,7V}{5k\Omega + 2k\Omega(1+100)} = 27,717 \mu A$$

$$I_{C2} = \beta \cdot I_{B2} = 100 \cdot 27,717 \mu A = 2,7717 mA$$

$$I_{E2} = I_{C2} + I_{B2} = 2,7995 mA$$

$$V_{C1} = V_{PLUS} - R_{C1} \cdot I_{RC1} = V_{PLUS} - R_{C1}(I_{C1} - I_{B2}) = 15V - 5k\Omega \cdot (1,2875 mA - 27,717 \mu A) = 8,7011 V$$

$$V_{E1} = R_{E1} \cdot I_{E1} = 3,9 V$$

$$V_{CE1} = V_{C1} - V_{E1} = 8,7011 V - 3,9 V = 4,8011 V > V_{CEsat} \Rightarrow \text{Transistor ist im aktiven Bereich}$$

$$V_{E2} = V_{PLUS} - R_{E2} \cdot I_{E2} = 15V - 2k\Omega \cdot 2,7995 mA = 9,401 V$$

$$V_{C2} = R_{C2} \cdot I_{C2} = 2,7k\Omega \cdot 2,7717 mA = 7,359 V$$

$$V_{EC2} = V_{E2} - V_{C2} = 9,401 V - 7,359 V = 2,0651 V > V_{ECsat} \Rightarrow \text{Transistor ist im aktiven Bereich}$$